

BUSINESS_PLAN.md & PRODUCT_PLAN.md - MASTER SPECIFICATION

Project Codename: Screen-Free AI Hardware Companion

Document Version: 1.0.0 (Master Execution Plan)

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1. Executive Summary & Vision

Our company is building a voice-first, screen-free interactive AI companion tailored for young children. By eliminating addictive screens and replacing them with warmth, low-latency conversational AI, and customized parent voice synthesis, we provide an engaging, safe, and educational companion for kids while giving parents complete control and peace of mind.

2. Hardware & Architecture Specifications

Component	Specification / Choice	Function / Purpose
Core Microcontroller	ESP32-S3 / Raspberry Pi Zero 2W	Wi-Fi/Bluetooth handling, local audio streaming, GPIO management
Audio Processing	I2S Microphone & Speaker Module	High-clarity voice capture and audio response playback
Cloud AI Engine	Custom AI Pipeline (LLM + TTS + STT)	Sub-second low-latency voice interaction and context

Component	Specification / Choice	Function / Purpose
		processing
Power Management	Li-ion Battery + USB-C Charging Module	Portable daily usage with integrated power safety circuits

3. Software Ecosystem & Parent Dashboard

The companion toy connects seamlessly to a mobile application designed for parents:

- **Voice Cloning & Customization:** Parents can record short voice samples to synthesize familiar, comforting tones for stories and conversations.
- **Activity & Safety Insights:** Real-time activity logs, educational progress tracking, and strict content filter controls.
- **Bedtime & Routine Schedules:** Automated routines for waking up, learning sessions, and peaceful bedtime story modes.

4. Go-To-Market (GTM) & Faceless "Build in Public" Strategy

Our strategy focuses on organic content-driven growth across social platforms while keeping the core founders focused on engineering and vision:

Platform	Target Audience	Content Strategy
TikTok / Instagram Reels	Parents, Early Tech Adopters	15-30s POV hardware assembly, voice interaction teasers, desk setups
LinkedIn	Investors, Tech Mentors, Talent	Founder updates, engineering breakthroughs, hardware cost optimization logs

Platform	Target Audience	Content Strategy
GitHub	Developer Community	Open-source landing pages, firmware documentation, architecture diagrams

5. 16-Week Execution Roadmap

- 1. **Month 1 (Aug - Sep 2026):** Development environment setup, baseline logic creation, and ESP32 audio streaming test.
- 2. **Month 2 (Sep - Oct 2026):** UI/UX wireframes for Parent App and HTML/CSS landing page deployment for pre-orders.
- 3. **Month 3 (Oct - Nov 2026):** AI voice pipeline integration, low-latency streaming optimization, and cloud backend setup.
- 4. **Month 4 (Nov - Dec 2026):** Full hardware prototype enclosure, beta testing with early users, and seed pitch preparation.